

Skill-6

1. To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

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ananya@DESKTOP-42I5F93:~$ mkdir -p skill6
ananya@DESKTOP-42I5F93:~$ cd skill6
ananya@DESKTOP-42I5F93:~/skill6$ pwd
/home/ananya/skill6
ananya@DESKTOP-42I5F93:~/skill6$ nano escape.c
ananya@DESKTOP-42I5F93:~/skill6$ gcc escape.c -o escape
ananya@DESKTOP-42I5F93:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world
Original : hello\ world
Parsed   : hello world
ananya@DESKTOP-42I5F93:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\&world
Original : hello\&world
Parsed   : hello&world
ananya@DESKTOP-42I5F93:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world\ \&\ test
Original : hello\ world\ \&\ test
Parsed   : hello world & test
ananya@DESKTOP-42I5F93:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\\world
Original : hello\\world
Parsed   : hello\world
ananya@DESKTOP-42I5F93:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world\ \&\ test\ \$100
Original : hello\ world\ \&\ test\ \$100
Parsed   : hello world & test $100
```

ananya@DESKTOP-42I5F93: ~ × + ▾

```
ananya@DESKTOP-42I5F93:~/skill6$ nano process.c
ananya@DESKTOP-42I5F93:~/skill6$ gcc process.c -o process
ananya@DESKTOP-42I5F93:~/skill6$ ./process ls
Parent Process
Parent PID: 770
Waiting for child process...
Child Process
Child PID: 771
Executing command: ls
escape escape.c process process.c
Child exited with status: 0
Parent process completed.
ananya@DESKTOP-42I5F93:~/skill6$ ./process echo Hello
Parent Process
Parent PID: 781
Waiting for child process...
Child Process
Child PID: 782
Executing command: echo
Hello
Child exited with status: 0
Parent process completed.
ananya@DESKTOP-42I5F93:~/skill6$ ./process pwd
Parent Process
Parent PID: 794
Waiting for child process...
Child Process
Child PID: 795
Executing command: pwd
/home/ananya/skill6
Child exited with status: 0
Parent process completed.
ananya@DESKTOP-42I5F93:~/skill6$ echo Hello
Hello
ananya@DESKTOP-42I5F93:~/skill6$ ./process pwd
Parent Process
Parent PID: 814
Waiting for child process...
Child Process
Child PID: 815
Executing command: pwd
```

```
/home/ananya/skill6
Child exited with status: 0
Parent process completed.
ananya@DESKTOP-42I5F93:~/skill6$ ./process abcxyz
Parent Process
Parent PID: 827
Waiting for child process...
Child Process
Child PID: 828
Executing command: abcxyz
execvp failed: No such file or directory
Child exited with status: 1
Parent process completed.
ananya@DESKTOP-42I5F93:~/skill6$ ./process date
Parent Process
Parent PID: 838
Waiting for child process...
Child Process
Child PID: 839
Executing command: date
Sat Sep 26 08:15:27 UTC 2026
Child exited with status: 0
Parent process completed.
ananya@DESKTOP-42I5F93:~/skill6$ ls
escape  escape.c  process  process.c
ananya@DESKTOP-42I5F93:~/skill6$ ./escape
Skill-6 Escape Sequence Parser
Enter input: hello\ world
Original : hello\ world
Parsed   : hello world
ananya@DESKTOP-42I5F93:~/skill6$ ./process ls
Parent Process
Parent PID: 888
Waiting for child process...
Child Process
Child PID: 889
Executing command: ls
escape  escape.c  process  process.c
Child exited with status: 0
Parent process completed.
```